SEPSIS SENTINEL
ICU EARLY-WARNING PLATFORM

RESEARCH / DEMO PROTOTYPE

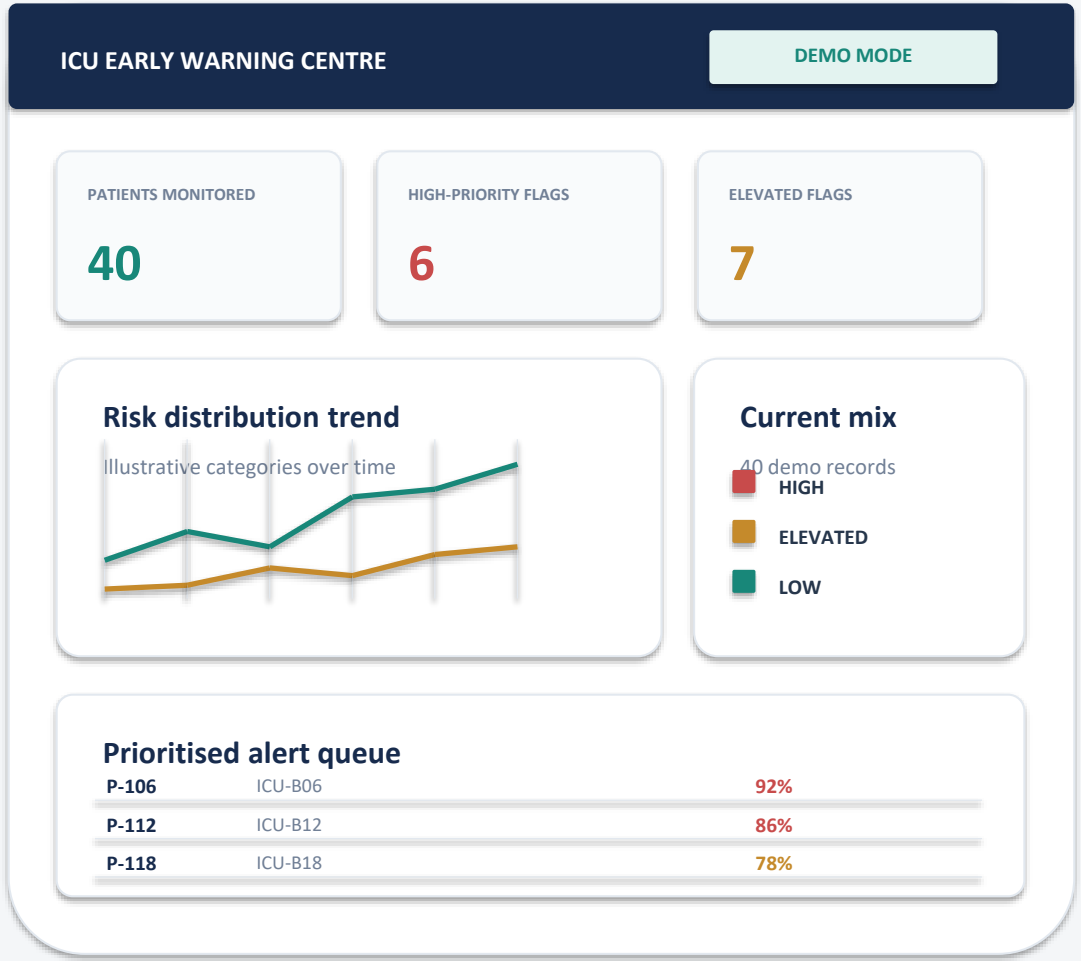
CLINICAL AI / CRITICAL CARE

EARLY SEPSIS PREDICTION

From ICU time-series data to a transparent, hourly risk summary for clinical review.

CORE IDEA

Spot changing patterns earlier — without replacing clinicians.



PRESENTED BY • Presenter Name

OCTOBER 2026

Synthetic values shown for interface demonstration only. No clinical validation is claimed.

01 / PROBLEM STATEMENT

The problem: sepsis is time-critical

Early deterioration can be subtle; ICU teams must interpret changing signals across many patients.

THE CHALLENGE

Signal is distributed across time, measurements and patients

A patient's condition can change before the overall pattern is obvious. A useful early-warning tool must surface changes early and clearly.

Subtle signals	Vitals and labs may shift gradually.
Irregular data	Measurements are missing or sampled at different times.
Alert fatigue	Too many false alarms can reduce trust.
Definition matters	Onset labels depend on the clinical definition and protocol.

WHY EARLY WARNING IS HARD

A credible system must be both timely and useful

01

Timing

Warnings must precede onset.

02

Data quality

Missingness and noise matter.

03

Clinical utility

Performance alone is not enough.

RESEARCH CONTEXT

Published estimates demonstrate the scale of the problem

48.9M

estimated cases worldwide in 2017

11.0M

estimated deaths worldwide in 2017

7.6%

lower survival per hour of antibiotic delay reported in septic shock study*

**Observational association; not a causal guarantee. Rudd et al. 2020; Kumar et al. 2006.*

02 / SOLUTION

Proposed solution: a transparent hourly risk workflow

The dashboard demonstrates how risk categories and patient observations could be organised for review.

01

CAPTURE

Hourly vitals, labs and context

Resample carefully; track missingness.

02

SUMMARISE

Feature windows and trends

A future trained model could estimate risk.

03

VALIDATE

Patient-level and external checks

Measure lead time, precision and utility.

04

PRESENT

Risk view with top drivers

Clinician reviews; tool does not decide.

DESIGN PRINCIPLE

A useful alert is early enough to review, explainable enough to question, and restrained enough to avoid overwhelming the care team.

EXPLAINABLE | TIME-AWARE | VALIDATION-FIRST | HUMAN-IN-LOOP

03 / TARGET USERS

Who the dashboard is designed to support

Different users need different views of the same underlying evidence.

01

ICU clinicians

Spot changes across beds

Ranked patient list, observations, trend context and visible demo-risk category.

02

Hospital quality & safety

Understand system-level behaviour

Alert volume, threshold sensitivity, subgroup checks and reporting.

03

Researchers & data teams

Build reproducible evidence

Dataset provenance, label definitions, patient-level splits and metrics.

ADOPTION DRIVERS

Synthetic-first prototyping • Explainable summaries • Workflow-aware alerts • Transparent validation status

04 / TECHNICAL ARCHITECTURE

Technical architecture: from observations to review

The current ZIP implements the dashboard layer; the model and clinical data pipeline remain future integration work.

INPUTS

Current interface fixture

Vitals
HR, SBP, RR, temp, SpO_2

Context
Patient ID, bed, pod

Completeness
Synthetic completeness field

Time series
Illustrative trend values

RISK / UI LOGIC

What the starter actually does

Heuristic score
Demo risk values in data.js

Configurable cutoffs
High and elevated sliders

Derived categories
HIGH / ELEVATED / LOW

Interactive views
Search, filters, patient selection

DASHBOARD

React + Vite + Recharts

Overview
KPIs and charts

Monitoring
Roster and trends

Risk analysis
Ranked demo alerts

Performance
Validation pending

Sources
Limits and provenance

FUTURE INTEGRATION

Not implemented in starter

PLANNED — **Permissioned dataset**
Document provenance

PLANNED — **Validated model**
Trained and versioned

PLANNED — **API / inference**
Schema and error checks

PLANNED — **Evaluation**
External and subgroup tests

05 / DATA & IMPLEMENTATION

Data and technology stack

The UI prototype uses local synthetic fixtures. Public ICU datasets are potential research sources, not connected live feeds.

DATA SOURCES

Potential research inputs

- MIMIC-IV
- eICU-CRD
- PhysioNet 2019

FEATURES

Planned processing

- Hourly windows
- Trend features
- Missingness masks

MODELS

Future candidates

- Gradient boosting
- LSTM / GRU
- Transformer

EVALUATION

Required evidence

- AUROC / AUPRC
- Lead time / PPV
- Calibration & utility

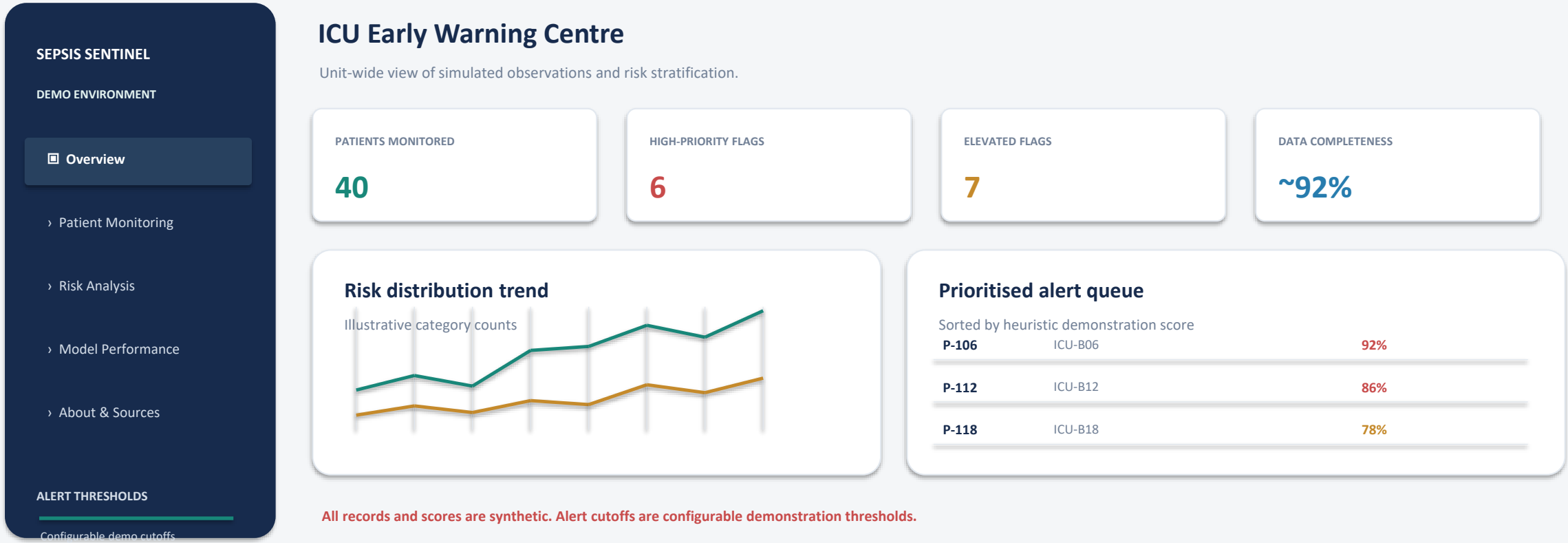
IMPLEMENTATION ROADMAP

01 Define labels and provenance → 02 Train baselines on patient-level splits → 03 Validate externally, calibrate and explain

06 / PRODUCT WALKTHROUGH

The Sepsis Sentinel dashboard

The dashboard is a research interface for exploring synthetic records—not a live clinical monitoring system.



07 / MODEL EVALUATION

Evaluation protocol: what must be proven

The starter intentionally reports no AUROC or AUPRC because it contains no trained, validated prediction model.

VALIDATION STATUS

P clinical performance claims

Clinical validation pending

Synthetic data and heuristic scores are for interface testing only.

- Define onset labels and prediction horizon
- Split by patient; test external site
- Check leakage and missingness
- Evaluate alert burden and subgroup safety

METRICS TO REPORT

Only after model training and validation

AUROC	AUPRC	Lead time	PPV / utility
Ranking quality	Positive-class precision/recall	Warning before onset	Actionable alerts vs burden

EVALUATION FLOW

ve protocol

1	2	3	4
DEFINE	SPLIT	SCORE	AUDIT
Sepsis-3 label	Patient + site	Hourly predictions	Safety + utility

08 / EVIDENCE & LIMITATIONS

External validation: performance may not transfer

A published evaluation of a widely implemented proprietary sepsis model illustrates why independent testing matters.

EPIC SEPSIS MODEL

Wong et al. · JAMA Internal Medicine, 2021

0.63
External AUC

38,455
hospitalisations

Reported sensitivity: 33%

67% of sepsis cases were missed

Alerts were generated for 18% of all patients at the studied threshold.

WHAT THIS MEANS

A model's published performance is not a guarantee in a new hospital.

Population shift
Patient mix and care pathways differ.

Workflow shift
Thresholds affect workload and response.

Independent testing
External validation is essential.

TAKEAWAY

Validation must assess more than a single headline score.

Measure missed cases, false alerts, calibration, lead time, subgroup performance and clinical workflow impact.

Source: Wong et al. (2021). Figures reflect the cited external validation study.

09 / RELATED WORK

Published case study: TREWScore and TREWS

Promising results are informative, but observational evidence should not be presented as proof of causality.

TREWScore · 2015

Henry et al. · Science Translational Medicine

EARLY WARNING

0.83
reported AUC

28.2 hours
median lead time reported

Machine learning on ICU data to predict septic shock.

TREWS in practice · 2022

Adams et al. · Nature Medicine

MULTI-SITE

18.7%
relative mortality reduction reported

≤ 3 hours
alert confirmation window

Lower mortality was associated with alerts confirmed within 3 hours.

CAUTION These findings are observational, not a randomised trial; results may not generalise to other settings.

10 / IMPLEMENTATION ROADMAP

Feasibility, risks and next steps

A dashboard can be prototyped on a laptop; clinical deployment requires a much higher evidence and governance bar.

FEASIBLE NOW

Prototype and research stage

✓ **Open tooling**	React, Vite, Recharts; Python model stack later
✓ **Synthetic-first UI**	Test navigation and visual hierarchy safely
✓ **Transparent limits**	No trained model metrics are fabricated
✓ **Human-in-the-loop**	Alerts support review, not diagnosis

RISKS → SAFEGUARDS

Required before any real-world use

Dataset shift	External validation and drift monitoring
Alert fatigue	Threshold analysis and PPV reporting
Label leakage	Define onset; remove post-onset inputs
Bias / trust	Subgroup audits and driver explanations

NEXT

Connect permissioned data → train baseline → external validation → silent-mode pilot → prospective evaluation

11 / REFERENCES

Research sources and responsible use

The following references underpin the research framing in the original deck.

Rudd et al. (2020) — BURDEN

Global, regional, and national sepsis incidence and mortality, 1990–2017. The Lancet.

Kumar et al. (2006) — TREATMENT DELAY

Duration of hypotension before effective antimicrobial therapy in septic shock. Critical Care Medicine.

Singer et al. (2016) — DEFINITION

The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA.

Reyna et al. (2020) — BENCHMARK

Early prediction of sepsis: the PhysioNet/Computing in Cardiology Challenge 2019. Critical Care Medicine.

Wong et al. (2021) — VALIDATION

External validation of a widely implemented proprietary sepsis prediction model. JAMA Internal Medicine.

Henry et al. (2015) · Adams et al. (2022) — CASE STUDY

TREWScore (Science Translational Medicine, 2015); prospective TREWS deployment (Nature Medicine, 2022).

IMPORTANT This is a research/demo interface using synthetic data and heuristic scores. It is not clinically validated and must not guide diagnosis, treatment or triage.